

Chapter- One

1. Introduction

At the end of this chapter students will be able to:

- ✓ Explain Biostatistics statistics
- ✓ identify types of level of measurement
- ✓ make a distinction between descriptive statistics and inferential statistics
- ✓ identify applications and limitations of Biostatistics

1.1 Definition and classification of biostatistics

Definition of Statistics

Statistics can be defined in two senses: plural (as Statistical Data) and singular (as Statistical Methods).

Statistical Data: Statistics are collection of facts (figures). This meaning of the word is widely used when reference is made to facts and figures on accident, weather, death, number of insects, etc. E.g.: Death Statistics, Labor Statistics, types of Insect statistics, etc. In this sense the word Statistics serves simply as data. But not all numerical data are statistics.

Statistical Methods: Statistics is the science that deals with the methods of data collection, organization, presentation, analysis and interpretation of data. It refers the subject area that is concerned with extracting relevant information from available data with the aim to make sound decisions.

Definition of biostatistics

Biostatistics is the scientific process of collecting, organizing, analyzing, presenting, interpreting and disseminating health and health-related data in estimating the magnitude of associations and testing hypotheses. Biostatistics can further be defined as a biological science and a branch of medicine that involves the processes of collection, analysis, interpretation, presentation of health and health-related information that may lead to conclusion and decision-making.

Classification of Statistics

The field of statistics is divided into two major divisions: *descriptive* and *inferential*.

Descriptive Statistics: refers to a discipline that quantitatively describes the important

characteristics of the dataset. For the purpose of describing different properties a data set, it uses measures of central tendency, i.e. mean, median, mode and the measures of dispersion i.e. range standard deviation, quartile deviation and variance, etc. Furthermore, the data is summarized by the researcher, in a useful way, with the help of numerical and graphical tools such as charts, tables, and graphs, to represent data in an accurate way. Moreover, the text is presented in support of the diagrams, to explain what they represent.

Inferential Statistics: is all about generalizing from the sample to the population, i.e. the results of analysis of the sample can be deduced to the larger population, from which the sample is taken. It is a convenient way to draw conclusions about the population when it is not possible to query each and every member of the universe. The sample chosen is a representative of the entire population; therefore, it should contain important features of the population.

For example, we may want to find the starting salary of a typical college graduate. To do so, we may select 2000 recent college graduates, find their starting salaries, and make a decision based on this information. The area of statistics that deals with such decision-making procedures is referred to as inferential statistics.

1.2 Stages in statistic investigation

According to Statistical Methods definition of statistics, a statistical study (statistical investigation) involves five stages:

1. **Collection of data:** the process of measuring, gathering, assembling the raw data up on which the statistical investigation is to be based. In this stage, you collect relevant data from multiple sources – both primary and secondary in nature.
2. **Organization of Data:** It is usually not possible to derive any conclusion about the main features of the data from direct inspection of the observations. This stage of statistical investigation helps to have a clear understanding of the information gathered and includes editing (correcting), classifying and tabulating the collected data in a systematic manner.
3. **Presentation of the data:** The process of re-organization, classification, compilation, and summarization of data to present it in a meaningful form. The mass data collected should

be presented in a suitable, concise form for further analysis. The collected data may be presented in the form of tabular or diagrammatic or graphic form.

4. **Analysis of data:** The process of extracting relevant information from the summarized data, mainly through the use of elementary mathematical operation. The data presented should be carefully analysis for making inference from the presented data such as measures of central tendencies, dispersion, correlation, regression etc.,
5. **Interpretation of data:** The interpretation and further observation of the various statistical measures through the analysis of the data by implementing those methods by which conclusions are formed and inferences made. The final step is drawing conclusion from the data collected. A valid conclusion must be drawn on the basis of analysis. A high degree of skill and experience is necessary for the interpretation.
 - Statistical techniques based on probability theory are required.

1.3 Definitions of some basic terms

Population: A population is a totality of things, objects, peoples, etc. about which information is being obtained. It is the largest collection of entities or values for which we have an interest at a particular time. Population can be finite (population consisting of fixed number of values) or infinite (population consisting of an endless succession of values).

Example: Patients of Heart Disease, Blood Pressure, etc.

Sampling: The process of selecting a sample from the population by some statistical method is called sampling.

Sample: A sample is a subset or part of a population which are selected by some statistical method to draw conclusions about the population.

Census survey: It is the process of examining the entire population. It is the total count of the population.

Parameter: It is a descriptive measure (value) computed from the population. It is the population measurement used to describe the population. Example: population mean and population standard deviation

Statistic: It is a measure (value) used to describe the sample and It is a value computed from the sample.

Sampling frame: A list of people, items or units from which the sample is taken.

Data: Data as a collection of related facts and figures from which conclusions may be drawn.

Variable: A certain characteristic which changes from object to object and time to time.

Sample size: The number of elements or observation to be included in the sample.

1.4 Applications, uses and limitation of biostatistics

Application of Biostatistics

The followings are some important applications and uses of Biostatistics as veritable tool in health and health-related areas:

- ❖ In Disease Surveillance.
- ❖ In Public Health epidemiology, health services, healthcare policy and management.
- ❖ In the evaluation of Healthcare services.
- ❖ In decision-making
- ❖ In prioritization of healthcare service implementation
- ❖ In comparison of health indices
- ❖ For the projection of health trends
- ❖ For the analysis of health and health-related data
- ❖ In Health Research e.g. in design and analysis of clinical and field trials in medicine
- ❖ In population genetics, statistical genetics in order to link variation in genotype with a variation in phenotype. This has been used in agriculture to improve crops and animal husbandry e.g. animal breeding.
- ❖ In ecological forecasting
- ❖ In biological sequence analysis.
- ❖ In systems biology for gene network inference or pathways analysis.

Use of Biostatistics:

- To test whether the difference between two population is real or by chance occurrence
- To study the correlation between attributes in the same population
- To evaluate the efficacy of vaccines
- To measure mortality and morbidity
- To evaluate the achievements of public health programs

- To fix priorities in public health programs

Limitations of statistics:

Statistics with all its wide application in every sphere of human activity has its own limitations. Some of them are given below.

- Statistics is not suitable to the study of qualitative phenomenon directly.
- Statistics does not study individuals.
- Statistical laws are not exact.
- Statistical results may be misused.
- Statistics is only one of the methods of studying a problem.

1.5 Types of variables and measurement scales

Qualitative variable: a variable or characteristic which cannot be measured in quantitative form but can only be identified by name or categories, for instance place of birth, ethnic group, type of drug, stages of breast cancer (I, II, III, or IV), degree of pain (minimal, moderate, severe or unbearable).

Quantitative variable: A quantitative variable is one that can be measured and expressed numerically and they can be of two types (discrete or continuous). The values of a discrete variable are usually whole numbers, such as the number of episodes of diarrhoea in the first five years of life. A continuous variable is a measurement on a continuous scale. Examples include weight, height, blood pressure, age, etc.

Types of Quantitative Variables:

- Discrete variable:** discrete variable occur when the observations are integers that correspond with a count of some sort. Some common examples are: the number of bacteria colonies on a plate, the number of cells within a prescribed area upon microscopic examination, the number of heart beats within a specified time interval, a mother's history of number of births (parity) and pregnancies (gravidity), the number of episodes of illness a patient experiences during some time period, etc.
- Continuous variable:** Each observation theoretically falls somewhere along a continuum. One is not restricted, in principle, to particular values such as the integers of the discrete scale. The restricting factor is the degree of accuracy of the measuring

instrument most clinical measurements, such as blood pressure, serum cholesterol level, height, weight, age etc. are on a continuous variable.

Example 1.3: Identify the qualitative/quantitative and discrete/continuous variables.

- a) Ron Steere sold 7 black cars and 8 red cars last month.
- b) Sara took her time when taking her final exam in a statistics class and got 85 correct answers in a multiple-choice test with 100 questions.

Solution

- a) Qualitative: Color of cars he sold.
Quantitative: The number of cars he sold.
- b) Discrete: Number of correct answers she got on the exam
Continuous: Length of time she took to finish the exam.

Type of Scales of measurement

Measurement is the assignment of numbers to objects or events in a systematic fashion. Four levels of measurement scales are commonly distinguished: nominal, ordinal, interval, and ratio and each possessed different properties of measurement systems.

Nominal Scale: Consists of ‘naming’ observations or classifying them into various mutually exclusive categories. Sometimes the variable under study is classified by some quality it possesses rather than by an amount or quantity. In such cases, the variable is called attribute. Example: Religion (Christianity, Islam, Hinduism, etc.), Sex (Male, Female), Eye color (blue, brown, black, etc.).

Ordinal Scale: With this scale observations are not only different from category to category, but can be ranked according to some criterion. The variables deal with their relative difference rather than with quantitative differences. Ordinal data are data which can have meaningful order. The inequality signs $<$ or $>$ may assume any meaning like ‘stronger, softer, weaker, better than’, etc.

Example:

- Patients may be characterized as unimproved, improved & much improved.
- Letter grading system i.e. A, B, C, D, F.
- Individuals may be classified according to socio-economic as low, medium & high.

Interval Scale: With this scale it is not only possible to order measurements, but also the distance between any two measurements is known but not meaningful quotients. There is no true zero point but arbitrary zero point. Interval data are the types of information in which an increase from one level to the next always reflects the same increase. Possible to add or subtract interval data but they may not be multiplied or divided. Example: Temperature of zero degrees does not indicate absence of the variable temperature. The two common temperature scales; Celsius (C) and Fahrenheit (F). We can see that the same difference exists between 10°C (50°F) and 20°C (68°F) as between 25°C (77°F) and 35°C (95°F) i.e. the measurement scale is composed of equal-sized interval. But we cannot say that a temperature of 20°C is twice as hot as a temperature of 10°C. because the zero point is arbitrary.

Ratio Scale: Characterized by the fact that equality of ratios as well as equality of intervals may be determined. Fundamental to ratio scales is a true zero point. We can do all basic arithmetic operations using variables measured by ratio scale. E.g.: Age, height, length, volume, etc.

Note: Variables measured by nominal and ordinal scale are qualitative variables.

1.5 Exercises

Identify the type of data (nominal, ordinal, interval and ratio) represented by each of the following.

1. Blood group
2. Temperature (Celsius)
3. Ethnic group
4. Job satisfaction index (1-5)
5. Number of heart attacks
6. Calendar year
7. Serum uric acid (mg/100ml)
8. Number of accidents in 3 - year period

Chapter-Two

Methods of Data Collection and Presentation

Introduction

Before any statistical work can be done data must be collected. Depending on the type of variable and the objective of the study different data collection methods can be employed.

Objectives: At the end of this chapter students will be able to:

- ✓ Arrange raw data in an array and then classified data to construct a frequency table and a cumulative frequency table.
- ✓ To organize data using frequency distribution.
- ✓ To present data using suitable graphs or diagrams

2.1 Methods of Data Collection

Data collection techniques allow us to systematically collect data about our objects of study (people, objects, and phenomena) and about the setting in which they occur. In the collection of data we have to be systematic. If data are collected haphazardly, it will be difficult to answer our research questions in a conclusive way. Decision-makers need information that is relevant, timely, accurate and usable. The cost of obtaining, processing and analyzing these data is high. The challenge is to find ways, which lead to information that is cost-effective, relevant, timely and important for immediate use. Some methods pay attention to timeliness and reduction in cost. Others pay attention to accuracy and the strength of the method in using scientific approaches.

2.2 Source of Data

The statistical data may be classified under two categories, depending upon the sources.

1) Primary data 2) Secondary data

Primary Data: are those data, which are collected by the investigator himself for the purpose of a specific inquiry or study. Such data are original in character and are mostly generated by surveys conducted by individuals or research institutions. The firsthand information obtained by the investigator is more reliable and accurate since the investigator can extract the correct information

by removing doubts, if any, in the minds of the respondents regarding certain questions. High response rates might be obtained since the answers to various questions are obtained on the spot. It permits explanation of questions concerning difficult subject matter.

There are many methods of collecting primary data and the main methods include:

Questionnaire: It is a popular means of collecting data, but is difficult to design and often require many rewrites before an acceptable questionnaire is produced.

Interviewing is a technique that is primarily used to gain an understanding of the underlying reasons and motivations for people's attitudes, preferences or behavior. Interviews can be undertaken on a personal one-to-one basis or in a group. They can be conducted at work, at home, in the street or in a shopping center, or some other agreed location.

Observation – Observation is a technique that involves systematically selecting, watching and recoding behaviors of people or other phenomena and aspects of the setting in which they occur, for the purpose of getting (gaining) specified information. It includes all methods from simple visual observations to the use of high level machines and measurements, sophisticated equipment or facilities, such as radiographic, biochemical, X-ray machines, microscope, clinical examinations, and microbiological examinations. Outline the guidelines for the observations prior to actual data collection.

Diaries: A diary is a way of gathering information about the way individuals spend their time on professional activities. They are not about records of engagements or personal journals of thought! Diaries can record either quantitative or qualitative data, and in management research can provide information about work patterns and activities.

Laboratory experiment: Conducting laboratory experiments on fields of chemical, biological sciences and so on.

Secondary Data: When an investigator uses data, which have already been collected by others, such data are called "Secondary Data". Such data are primary data for the agency that collected them, and become secondary for someone else who uses these data for his own purposes. The secondary data can be obtained from journals, reports, government publications, publications of professionals and research organizations. Secondary data are less expensive to collect both in money and time. These data can also be better utilized and sometimes the quality of such data may be better because these might have been collected by persons who were specially trained for that purpose. On the other hand, such data must be

used with great care, because such data may also be **full of errors** due to the fact that the purpose of the collection of the data by the primary agency may have been different from the purpose of the user of these secondary data. Secondly, there may have been bias introduced, the size of the sample may have been inadequate, or there may have been arithmetic or definition errors, hence, it is necessary to critically investigate the validity of the secondary data.

Secondary data analysis can be literally defined as second-hand analysis and is the analysis of data or information that was either gathered by someone else (e.g., researchers, institutions, other NGOs, etc.) or for some other purpose than the one currently being considered, or often a combination of the two.

Some of the sources of secondary data are government document, official statistics, technical report, scholarly journals, trade journals, review articles, reference books, research institutes, universities, hospitals, libraries, library search engines, computerized data base and world wide

2.3 Methods of data presentation

The data collected in a survey is called raw data. In most cases, useful information is not immediately evident from the mass of these raw data. Collected data need to be organized in such a way as to condense the information they contain in a way that will show patterns of variation clearly. Therefore, for the raw data to be more easily appreciated and to draw quick comparisons, it is often useful to present the data in the form of:

Having collected and edited the data, the next important step is to organize it. That is to present it in a readily comprehensible condensed form that aids in order to draw inferences from it. It is also necessary that the like be separated from the unlike ones.

The presentation of data is broadly classified in to the following two categories:

- Tabular presentation
- Diagrammatic and Graphic presentation

Classification is a preliminary and it prepares the ground for proper presentation of data.

Definitions:

- Raw data: recorded information in its original collected form, whether it is counts or measurements, is referred to as raw data.

- Frequency: is the number of values in a specific class of the distribution.
- Frequency distribution: is the organization of raw data in table form using classes and frequencies.

There are three basic types of frequency distributions

- Categorical frequency distribution
- Ungrouped frequency distribution
- Grouped frequency distribution

There are specific procedures for constructing each type.

i. Categorical frequency Distribution:

Used for data that can be place in specific categories such as nominal, or ordinal.

E.g: consider the immunization Status of children in Bahir Dar town

NI, PI, FI, NI, PI, FI, NI, PI, FI, NI, PI, FI, NI, PI, FI, NI, NI, NI, NI, NI, NI, PI, PI, PI, PI, PI, PI, FI, FI, FI, FI, FI, FI, FI, FI, FI, FI, FI, FI, PI, NI, NI.

Where NI denotes not immunized children, PI denotes partially immunized children, FI fully immunized children

Construct Frequency distribution.

Solution:

Since the data are categorical, discrete classes can be used. There are three types of Immunization Status NI, PI and FI. These types will be used as class for the distribution. We follow procedure to construct the frequency distribution.

$$\% = \frac{f}{n} * 100 \quad \text{Where } f = \text{frequency of the class, } n = \text{total number of value.}$$

Table 1: Overall Immunization Status of children in Bahir Dar city

Immunization Status	Tally	Frequency	Percent
Not Immunized	/// //X ///	13	32.5
Partially Immunized	/// //X //	12	30
Fully Immunized	/// //X //X	15	37.5
Total		40	100.0

ii. Ungrouped frequency Distribution:

-Is a table of all the potential raw score values that could possible occur in the data along with the number of times each actually occurred.

-Is often constructed for small set or data on discrete variable.

Constructing ungrouped frequency distribution:

- First find the smallest and largest raw score in the collected data.
- Arrange the data in order of magnitude and count the frequency.
- To facilitate counting one may include a column of tallies.

Example: The following data represent the mark of 10 students.

80 76 90 85 80
70 60 62 70 85

Construct a frequency distribution, which is ungrouped.

Solution:

Step 1: Find the range, Range=Max-Min=90-60=30.

Step 2: Make a table as shown

Step 3: Tally the data.

Step 4: Compute the frequency.

Mark	Tally	Frequency
60	//	1
62	/	1
70	////	2
76	/	1
80	///	2
85	///	2
90	/	1

Each individual value is presented separately, that is why it is named ungrouped frequency distribution.

iii. **Grouped frequency Distribution:** When the range of the data is large, the data must be grouped in to classes that are more than one unit in width.

Definitions:

- **Grouped Frequency Distribution:** a frequency distribution when several numbers are grouped in one class.
- **Class limits:** Separates one class in a grouped frequency distribution from another. The limits could actually appear in the data and have gaps between the upper limits of one class and lower limit of the next.
- **Units of measurement (U):** the distance between two possible consecutive measures. It is usually taken as 1, 0.1, 0.01, 0.001, -----.
- **Class boundaries:** Separates one class in a grouped frequency distribution from another. The boundaries have one more decimal places than the row data and therefore do not appear in the data. There is no gap between the upper boundary of one class and lower boundary of the next class. The lower class boundary is found by subtracting $U/2$ from the corresponding lower class limit and the upper class boundary is found by adding $U/2$ to the corresponding upper class limit.
- **Class width:** the difference between the upper and lower class boundaries of any class. It is also the difference between the lower limits of any two consecutive classes or the difference between any two consecutive class marks.
- **Class mark (Mid points):** it is the average of the lower and upper class limits or the average of upper and lower class boundary.
- **Cumulative frequency:** is the number of observations less than/more than or equal to a specific value.
- **Cumulative frequency above:** it is the total frequency of all values greater than or equal to the lower class boundary of a given class.
- **Cumulative frequency below:** it is the total frequency of all values less than or equal to the upper-class boundary of a given class.
- **Cumulative Frequency Distribution (CFD):** it is the tabular arrangement of class interval together with their corresponding cumulative frequencies. It can be more than or less than type, depending on the type of cumulative frequency used.
- **Relative frequency (RF):** it is the frequency divided by the total frequency.
- **Relative cumulative frequency (RCF):** it is the cumulative frequency divided by the total frequency.

Steps for constructing Grouped frequency Distribution

1. Find the largest and smallest values
2. Compute the Range(R) = Maximum - Minimum
3. Select the number of classes desired, usually between 5 and 20 or use Sturges rule $k = 1 + 3.322 \log n$ where k is number of classes desired and n is total number of observation.
4. Find the class width by dividing the range by the number of classes and rounding up, not off. $w = \frac{R}{k}$.
5. Pick a suitable starting point less than or equal to the minimum value. The starting point is called the lower limit of the first class. Continue to add the class width to this lower limit to get the rest of the lower limits.
6. To find the upper limit of the first class, subtract U from the lower limit of the second class. Then continue to add the class width to this upper limit to find the rest of the upper limits.
7. Find the boundaries by subtracting U/2 units from the lower limits and adding U/2 units from the upper limits. The boundaries are also half-way between the upper limit of one class and the lower limit of the next class. It may not be necessary to find the boundaries.
8. Tally the data.
9. Find the frequencies.
10. Find the cumulative frequencies. Depending on what you're trying to accomplish, it may not be necessary to find the cumulative frequencies.
11. If necessary, find the relative frequencies and/or relative cumulative frequencies

Example

Example 2.3: Construct FD for the following data.

11 29 33 22 27 19 22 21 18 17 22 38 26 39 27 6 34 13 20

Solution: -

- 1) Highest value = 39, Lowest value = 6
- 2) Range = $39 - 6 = 33$
- 3) $K = 1 + 3.322 \log 20 = 1 + 3.322(2.301) = 5.6 \approx 6$
- 4) $W = R / K = 33/6 = 5.5 \approx 6$
- 5) $U = 1$

6) $LCL_1 = 6$

7) Find the upper-class limits. Find class boundaries, find class mark and tally the data

Class limit	Class boundary	Class Mark	Tally	Frequency	CF(<)	CF(>)	RF	RCF(>)
6 – 11	5.5 – 11.5	8.5	//	2	2	20	$2/20=0.1$	1
12 – 17	11.5 – 17.5	14.5	//	2	4	18	$2/20=0.1$	0.9
18 – 23	17.5 – 23.5	20.5	///// //	7	11	16	$7/20=0.35$	0.8
24 – 29	23.5 – 29.5	26.5	////	4	15	9	$4/20=0.2$	0.45
30 – 35	29.5 – 35.5	32.5	///	3	18	5	$3/20=0.15$	0.25
36 – 41	35.5 – 41.5	38.5	//	2	20	2	$2/20=0.1$	0.10

The above frequency distribution shows the way in which the values of the variable are distributed among the specified class intervals

2.1.1. Diagrammatic presentation of data.

These are techniques for presenting data in visual displays using geometric and pictures.

Importance:

- They have greater attraction.
- They facilitate comparison.
- They are easily understandable.

Diagrams are appropriate for presenting discrete or categorical data. The three most commonly used diagrammatic presentation for discrete as well as qualitative data are:

- Pie charts
- Pictogram
- Bar charts

Pie chart

Pie charts are another popular type of graph. In applications suitable for pie charts, there is only one group but we want to decompose it into several categories. A pie chart consists of a circle; the circle is divided into wedges that correspond to the magnitude of the proportions for various categories. The angle of the sector is obtained using:

$$\text{Angle of sector} = \frac{\text{Value of the part}}{\text{the whole quantity}} * 360$$

Example: Draw a suitable diagram to represent the following population in a town.

Immunization Status	Tally	Frequency	Percent
Not Immunized		13	32.5
Partially Immunized		12	30
Fully Immunized		15	37.5

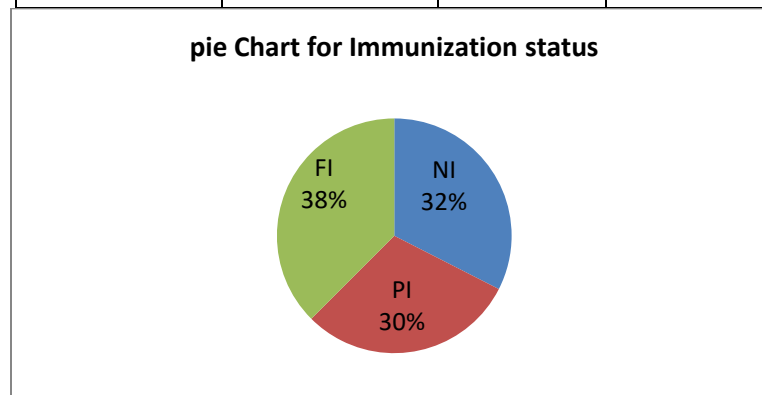
Solutions:

Step 1: Find the percentage.

Step 2: Find the number of degrees for each class.

Step 3: Using a protractor and compass, graph each section and write its name corresponding percentage.

Class	Frequency	Percent	Degree
NI	13	32.5	117
PI	12	30	108
FI	15	37.5	135



Pictogram

In this diagram, we represent data by means of some picture symbols. We decide about a suitable picture to represent a definite number of units in which the variable is measured.

Example: draw a pictogram to represent the following population of a town.

Year	1989	1990	1991	1992
Population	2000	3000	5000	7000

Bar Chart

Bar charts are a very popular type of graph used to display several proportions for quick comparison. In applications suitable for bar charts, there are several groups and we investigate one binary characteristic. In a bar chart, the various groups are represented along the horizontal axis; they may be arranged alphabetically, by the size of their proportions, or on some other rational basis. The bars should be of equal width and should be separated from one another so as not to imply continuity. A set of bars (thick lines or narrow rectangles) representing some magnitude over time space. Bars can be drawn either vertically or horizontally. A vertical bar is drawn above each group such that the height of the bar is the proportion associated with that group.

There are different types of bar charts. The most common being:

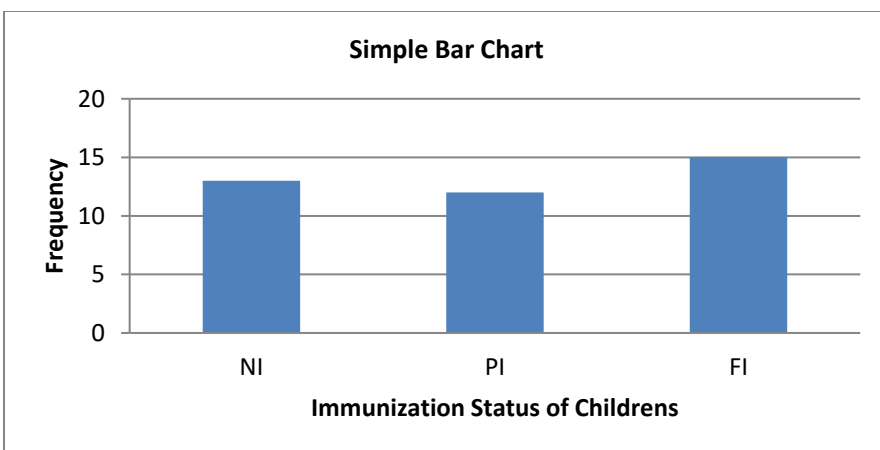
- Simple bar chart
- Component or sub divided bar chart.
- Multiple bar charts.

Simple Bar Chart: It is used to display data on one variable. They are thick lines (narrow rectangles) having the same breadth. The magnitude of a quantity is represented by the height /length of the bar.

Example: consider immunization status of children data, construct simple bar chart.

Immunization Status	Tally	Frequency	Percent
Not Immunized		13	32.5
Partially Immunized		12	30
Fully Immunized		15	37.5

Solutions:



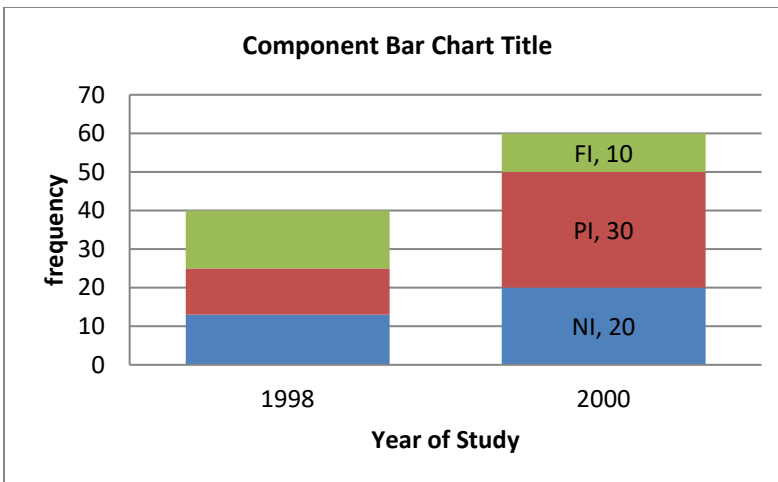
Component Bar chart: When there is a desire to show how a total (or aggregate) is divided in to its component parts, we use component bar chart. The bars represent total value of a variable with each total broken in to its component parts and different colors or designs are used for identifications.

Example:

Draw a component bar chart to represent the immunization status of children from 1998 to 2000.

Year of study	1998	2000
Immunization Status	Frequency	Frequency
Not Immunized	13	20
Partially Immunized	12	30
Fully Immunized	15	10

Solutions:

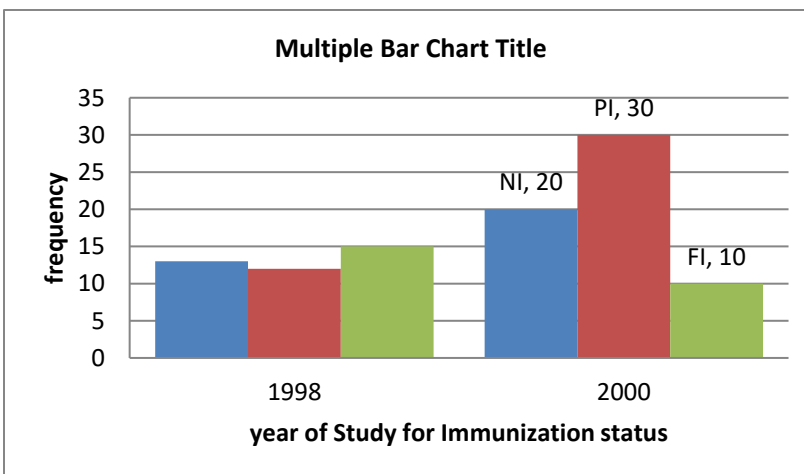


Multiple Bar charts: These are used to display data on more than one variable. They are used for comparing different variables at the same time.

Example:

Draw a multiple bar chart to represent the immunization status from 1998 & 2000 from the above example.

Solutions:



2.1.2. Graphical Presentation of data

The histogram, frequency polygon and cumulative frequency graph or ogive is most commonly applied graphical representation for continuous data.

Procedures for constructing statistical graphs:

- Draw and label the X and Y axes.

- Choose a suitable scale for the frequencies or cumulative frequencies and label it on the Y axes.
- Represent the class boundaries for the histogram or ogive or the mid points for the frequency polygon on the X axes.
- Plot the points.
- Draw the bars or lines to connect the points.

Histogram

A convenient way of displaying a frequency table is by means of a histogram and/or a frequency polygon. It is a graph which displays the data by using vertical bars of various heights to represent frequencies. A histogram is a graph in which:

1. The horizontal scale represents the value of the variable marked at interval boundaries.
2. The vertical scale represents the frequency or relative frequency in each interval.

A histogram presents us with a graphic picture of the distribution of measurements. This picture consists of rectangular bars joining each other.

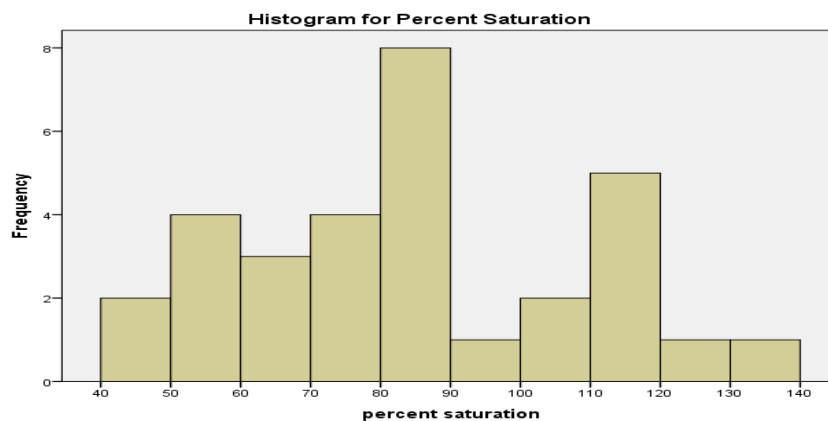
Frequency Polygon: To draw a frequency polygon, we first place a dot at the midpoint of the upper base of each rectangular bar. The points are connected with straight lines. At the ends, the points are connected to the midpoints of the previous and succeeding intervals (these are make-up intervals with zero frequency, where widths are the widths of the first and last intervals, respectively). A frequency polygon so constructed is another way to portray graphically the distribution of a data set. The frequency polygon can also be shown without the histogram on the same graph.

Example 2.4 Table 2.4 provides data on age and percentage saturation of bile for 31 male patients. Using 10% intervals, the data set can be represented by a histogram or a frequency polygon as shown in Graph 2.4 & 2.5. This graph shows that among the nine patients with over 100% saturation, eight (or 89%) are over 50 years of age. On the other hand, only four of 22 (or 18%) patients with less than 100% saturation are over 50 years of age. The two peaks in Figure 2.4 might correspond to the two age groups.

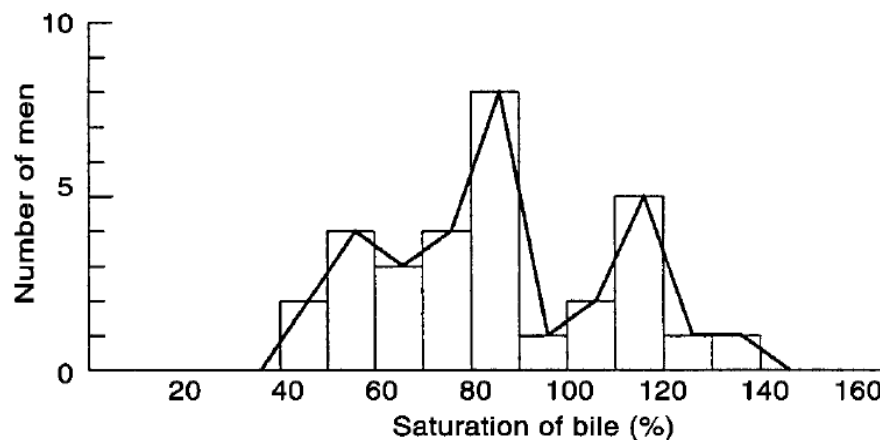
Table: 2.4 Percent Saturation bile

Age	Percent Saturation	Age	Percent Saturation	Age	Percent Saturation
23	40	55	137	48	78
31	86	31	88	27	80
58	111	20	88	32	47
25	86	23	65	62	74
63	106	43	79	36	58
43	66	27	87	29	88
67	123	63	56	27	73
48	90	59	110	65	118
29	112	53	106	42	67
26	52	66	110	60	57
64	88				

Graph 2.4 histogram of percent saturation bile



Graph 2.5 Frequency polygons with Histogram

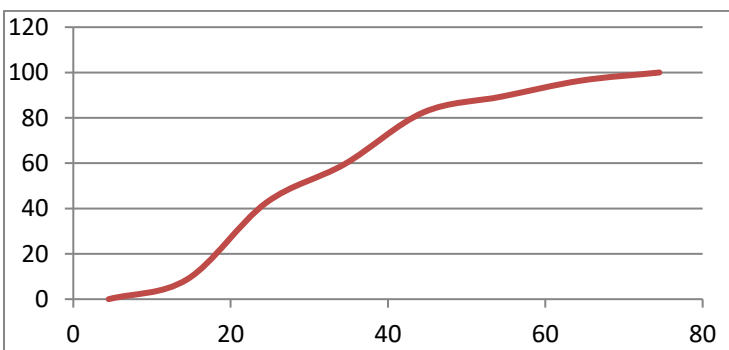


Another application concerns the symmetry of the distribution as depicted by the table or its graphs. A symmetric distribution is one in which the distribution has the same shape on both sides of the peak location.

Cumulative Frequency Graph (Ogive)

Cumulative relative frequency, or ogive, gives the percentage of persons having a measurement less than or equal to the upper boundary of the class interval.

Weight Interval (lb)	Frequency	Relative Frequency (%)	Cumulative Relative Frequency (%)
10–19	5	8.8	8.8
20–29	19	33.3	42.1
30–39	10	17.5	59.6
40–49	13	22.8	82.4
50–59	4	7.0	89.4
60–69	4	7.0	96.4
70–79	2	3.5	99.9 $\approx$ 100.0
Total	57	100.0	



The cumulative frequency graph, have a number of applications. When two cumulative frequency graphs, representing two different data sets, are placed on the same graph, they provide a rapid visual comparison without any need to compare individual intervals.

Chapter-3

3. Measures of Central Tendency

The tendency of statistical data to get concentrated at certain values is called the “Central Tendency” and the various methods of determining the actual value at which the data tend to concentrate are called measures of central Tendency or averages. Hence, an average is a value which tends to sum up or describe the mass of the data.

3.1 Objectives of measures of central tendency

- It facilitates quick understanding of complex data.
- It facilitates comparison
- To know about the universe from the sample
- To get the single value that describes the characteristic of the entire group.

3.2 Summation notation

Statistical Symbols: Let a data set consists of a number of observations, represents by $x_1, x_2, \dots, x_n$ where n (the last subscript) denotes the number of observations in the data and x_i is the i^{th} observation. Then the sum of all numbers (x_i 's) where i goes from 1 up to n is symbolically given by $\sum_{i=1}^n x_i$ or $\sum x_i$ or $\sum x$ that is,

$$\sum x_i = x_1 + x_2 + \dots + x_n$$

x - Whole set of numbers

x_i - Specific score in a set of numbers

n - Total number of observations

The Greek letter $\sum$ is the summation sign

First, let us suppose that we have n measurements in a data set; for example, here is a data set: 8, 2, 3, and 5 with $n = 4$. We usually denote these numbers as x_i 's; thus we have for the example above: $x_1 = 8$, $x_2 = 2$, $x_3 = 3$, and $x_4 = 5$. If we add all the x_i 's in the data set above, we obtain 18 as the sum. This addition process is recorded as $\sum x_i = 18$. With the summation notation, we are now able to define a number of important summarized measures, starting with the arithmetic average or mean.

Some Properties of the Summation Notation

1. $\sum_{i=1}^n c = n.c$, where c is a constant number.
2. $\sum_{i=1}^n bx_i = b\sum_{i=1}^n x_i$ where b is a constant number
3. $\sum_{i=1}^n (a + bx_i) = n.a + b\sum_{i=1}^n x_i$
4. $\sum_{i=1}^n ((x_i \pm y_i)) = \sum_{i=1}^n x_i \pm \sum_{i=1}^n y_i$
5. $\sum_{i=1}^n x_i y_i \neq \sum_{i=1}^n x_i \sum_{i=1}^n y_i$

3.3 Important characteristics of good average

The following properties should possess for a good average.

1. It should be rigidly defined.
2. It should be easy to understand and compute.
3. It should be based on all items in the data.
4. Its definition shall be in the form of a mathematical formula.
5. It should be capable of further algebraic treatment.
6. It should have sampling stability.
7. It should be capable of being used in further statistical computations or processing.

3.4 Mean**3.4.1 Arithmetic mean**

Arithmetic means defined as the sum of the measurements of the items divided by the total number of items. It is usually denoted by $\bar{x}$ ("x-bar").

Given a data set of size n , $x_1, x_2, \dots, x_n$ the mean of the x 's will be denoted by $\bar{x}$ ("x-bar") and is computed by summing all the x 's and dividing the sum by n .

Symbolically,

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

It is important to know that $\sum$ ("sigma") stands for an operation (that of obtaining the sum of the quantities that follow) rather than a quantity itself.

Example: Below are the measured lengths of 30 individual bacterial cells in μm . As they have not yet been sorted to make a sorted list, we want to calculate the average length of bacterial cell. Find it??

1.5 , 2.0 , 2.0 , 3.0 , 2.0 , 3.2 , 2.3 , 1.5 , 2.0 , 2.0 , 1.0 , 1.0 , 2.5 , 3.4 , 2.1 , 2.0 , 4.0 , 3.0 , 2.0 , 2.0 , 2.2 , 2.0 , 2.0 , 2.0 , 2.0 , 1.5 , 2.0 , 1.0 , 1.0 , 1.0

Solution:

To find the average value of bacterial length we apply the above arithmetic mean formula.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{x} = \frac{1.5 + 2.0 + 2.0 + \dots + 1.0}{30}$$

$$\bar{x} = 2.6733$$

The average length of bacteria is 2.673µm.

Arithmetic mean for grouped data

Occasionally, data, especially secondhand data, are presented in the grouped form of a frequency table. In these cases, the mean $\bar{x}$ can be approximated using the formula

$$\bar{x} = \frac{\sum_{i=1}^k x_i f_i}{\sum_{i=1}^k f_i} = \frac{x_1 f_1 + x_2 f_2 + \dots + x_k f_k}{f_1 + f_2 + \dots + f_k}$$

Where f_i denotes the frequency (i.e., the number of observations in an interval), x_i the interval midpoint or the value of $\bar{x}$'s when the data are in the form of ungrouped frequency distribution, and the summation is across the intervals. The midpoint for an interval is obtained by calculating the average of the interval lower true boundary and the upper true boundary.

Example, suppose that we measure the weight of student in a certain school and the results are presented in table 3.1. Calculate the average of weight of students based on the table 3.1.

Table 3.1 Weight of students in lb

Weight interval	Frequency
10 – 19	5
20 – 29	19
30 – 39	10
40 – 49	13
50 – 59	4

60 – 69	4
70 – 79	2

Solution:

If the first three intervals are

10–19

20–29

30–39

The midpoint for the first interval is

$$\frac{9.5 + 19.5}{2} = 14.5$$

And for the second interval is

$$\frac{19.5 + 29.5}{2} = 24.5$$

This process for calculation of the mean $\bar{x}$ is illustrated in the following Table. 7

Weight interval	Midpoint(m)	Frequency
10 – 19	14.5	5
20 – 29	24.5	19
30 – 39	34.5	10
40 – 49	44.5	13
50 – 59	54.5	4
60 – 69	64.5	4
70 – 79	74.5	2

Then the average weight of students is:

$$\bar{x} = \frac{\sum fm}{n} = \frac{2086.5}{57} = 36:7\text{lb.}$$

Of course, the mean $\bar{x}$ obtained from this technique with a frequency table is different from the $\bar{x}$ using individual or raw data. However, the process saves some computational labor, and the difference between the results, $\bar{x}$'s, is very small if the data set is large and the interval width is small.

Properties of the Arithmetic Mean

- The algebraic sum of the deviations of a set of numbers $x_1, x_2, \dots, x_n$ from their mean $\bar{x}$ is always zero. i.e.

$$\sum_{i=1}^n (x_i - \bar{x}) = 0$$

- The sum of squares of deviations from the mean is the least. That is, $\sum_{i=1}^n (x_i - A)^2$ is minimum when $A = \bar{x}$.

- If the mean of $x_1, x_2, \dots, x_n$ is $\bar{x}$, then
 - a) The mean of $x_1 \pm k, x_2 \pm k, \dots, x_n \pm k$ will be $\bar{x} \pm k$
 - b) The mean of $kx_1, kx_2, \dots, kx_n$ will be $k\bar{x}$.

Merits of Arithmetic Mean

- Arithmetic mean has a rigidly defined mathematical formula so that its value is always definite or unique. It can be calculated for any set of numerical data.
- It is calculated based on all observations.
- Arithmetic mean is simple to calculate and easy to understand.
- It doesn't need arrangement of data in increasing or decreasing order.
- Arithmetic mean of many samples from the same population does not fluctuate considerably.
- It affords a good standard of comparison.

Demerits of Arithmetic Mean

- It can't be calculated for data which are not quantifiable.
- It is highly affected by extreme (abnormal) values in the series.
- It can be a number which does not exist in the series.
- It can't be calculated for grouped continuous open-ended classes.

3.4.1.2 Weighted Arithmetic Mean

While calculating simple arithmetic mean, all items were assumed to be of equal importance (each value in the data set has equal weight). When the observations have different weight, we use weighted average. Weights are assigned to each item in proportion to its relative importance.

If $x_1, x_2, \dots, x_n$ represent values of the items and $w_1, w_2, \dots, w_n$ are the corresponding weights, then the weighted mean, $(\bar{x}_w)$ is given by

$$\bar{x}_w = \frac{w_1x_1 + w_2x_2 + \dots + w_nx_n}{w_1 + w_2 + \dots + w_n} = \frac{\sum w_i x_i}{\sum w_i}$$

Example: A student's final mark in Mathematics, Physics, Chemistry and Biology are respectively A, B, D and C. If the respective credits received for these courses are 4, 4, 3 and 2, determine the approximate average grade the student has got for the course.

Solution

We use a weighted arithmetic mean, weight associated with each course being taken as the number of credits received for the corresponding course.

x_i	4	3	1	2	Total
w_i	4	4	3	2	13
$x_i w_i$	16	12	3	4	35

$$\bar{x}_w = \frac{w_1x_1 + w_2x_2 + \dots + w_nx_n}{w_1 + w_2 + \dots + w_n} = \frac{\sum w_i x_i}{\sum w_i}$$

$$= \frac{16+12+3+4}{13} = \frac{35}{13} = 2.69$$

Average grade of the student is approximately 2.69.

Combined mean: When a set of observations is divided into k groups and $\bar{x}_1$ is the mean of n_1 observations of group 1, $\bar{x}_2$ is the mean of n_2 observations of group 2, ..., $\bar{x}_k$ is the mean of n_k observations of group k , then the combined mean, denoted by $\bar{x}_c$, of all observations taken together is given by

$$\bar{x}_c = \frac{\bar{x}_1 n_1 + \bar{x}_2 n_2 + \cdots + \bar{x}_k n_k}{n_1 + n_2 + \cdots + n_k}$$

This is a special case of the weighted mean. In this case the sample sizes are the weights.

Example: In the Previous year there were two sections taking Statistics course. At the end of the semester, the two sections got average marks of 70 & 78. There were 45 and 50 students in each section respectively. Find the mean mark for the entire students.

Solution:

$$\bar{x}_c = \frac{\bar{x}_1 n_1 + \bar{x}_2 n_2 + \cdots + \bar{x}_k n_k}{n_1 + n_2 + \cdots + n_k} = \frac{\bar{x}_1 n_1 + \bar{x}_2 n_2}{n_1 + n_2} = \frac{70 \times 45 + 78 \times 50}{45 + 50} = \frac{7050}{95} = 74.21$$

The combined mean of the entire students will be 74.21.

3.4.1.3 Geometric Mean

The geometric mean is preferable to the arithmetic mean if the series of observations contains one or more unusually large values. The above method of calculating geometric mean is satisfactory only if there are a small number of items. But if n is a large number, the problem of computing the n^{th} root of the product of these values by simple arithmetic is a tedious work.

To facilitate the computation of geometric mean we make use of logarithms.

As indicated earlier, a characteristic of some interest is the symmetry or lack of symmetry of a distribution, and it is recommended that for very positively skewed distributions, analyses are commonly done on the log scale. After obtaining a mean on the log scale, we should take the antilog to return to the original scale of measurement; the result is called the geometric mean of the x's.

The effect of this process is to minimize the influences of extreme observations (very large numbers in the data set). For example, considering the data set 8, 5, 4, 12, 15, 7, 28 with one unusually large measurement, we have Table 3.2, with natural logs presented in the second column.

Table 3.2

x	$\ln x$
8	2.08
5	1.61
4	1.39
12	2.48
15	2.71
7	1.95
28	3.33
79	15.55

The mean is

$$\bar{x} = \frac{79}{7} = 11.3$$

While on the log scale we have

$$\bar{x} = \frac{\sum \ln x}{n} = \frac{15.55}{7} = 2.22$$

Leading to a geometric mean of 9.22, which is less affected by the large or small measurements?

$G.M = \sqrt[n]{x_1 \cdot x_2 \dots x_n} = \text{antilog} \left(\frac{1}{n} \sum \log x_i \right)$ when the data are presented in series form.

When the numbers $x_1, x_2, \dots, x_k$ occur with frequencies $f_1, f_2, \dots, f_m$ respectively, then the geometric mean is obtained by

$$G.M. = \sqrt[n]{x_1^{f_1} \cdot x_2^{f_2} \dots x_m^{f_m}} = \text{antilog} \left(\frac{1}{n} \sum f_i \log x_i \right)$$

Where x 's are midpoint if the data are presented in the form of grouped frequency distribution or it is the value of x 's if the data are presented in the form of frequency distribution and f_i 's are the frequency of each classes. Geometric mean is used extensively in microbiological and serological research, in which distributions are often skewed positively.

Example:

The remission times of 42 patients with acute leukemia were reported from a clinical trial undertaken to assess the ability of the drug 6-mercaptopurine (6-MP) to maintain remission. Each patient was randomized to receive either 6-MP or placebo. The study was terminated after one year; patients have different follow-up times because they were enrolled sequentially

at different times. Times to relapse in weeks for the 21 patients in the placebo group were 1, 1, 2, 2, 3, 4, 4, 5, 5, 8, 8, 8, 8, 11, 11, 12, 12, 15, 17, 22, and 23

The Geometric mean is on the *log* scale we have

$$\ln(G.M) = \frac{\sum_{i=1}^{21} f_i \ln x_i}{n} = \frac{2 \ln(1) + 2 \ln(2) + \dots + \ln(23)}{21} = 1.826$$

$$G.M(\bar{x}) = \text{antilog}(1.826)$$

Leading to a geometric mean of 6.21, which, in general, is less affected by the large measurements?

3.4.1.4 Harmonic Mean

It is a suitable measure of central tendency when the data pertains to speed, rate and time. The harmonic of n values is defined as n divided by the sum of their reciprocal.

Harmonic mean for individual series: If $x_1, x_2, \dots, x_n$ are n observations, then harmonic mean can be represented by the following formula:

$$H.M = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$$

Example 3.9 A car travels 25 miles at 25 mph, 25 miles at 50 mph, and 25 miles at 75 mph. Find the harmonic mean of the three velocities.

Solution

$$H.M = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{3}{\frac{1}{25} + \frac{1}{50} + \frac{1}{75}} = 40.9$$

Harmonic mean for discrete data arranged in FD: If the data is arranged in the form of frequency distribution

$$H.M = \frac{n}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \dots + \frac{f_m}{x_m}}, \text{ where } n = \sum_{k=1}^m f_k$$

Harmonic mean for continuous grouped FD: Whenever the frequency distribution are grouped continuous, class marks of the class intervals are considered as x_i and the above formula can be used as

$$\text{H.M.} = \frac{n}{\sum_{i=1}^n \frac{f_i}{x_i}} \text{ where } n = \sum_{k=1}^m f_k$$

x_i is the class mark of i^{th} class

Properties of harmonic mean

- It is unique for a given set of data.
- It takes each and every observation into consideration.
- Difficult to calculate and understand.
- Appropriate measure of central tendency in situations where data is in ratio, speed or rate.

Relations among different means

i. If all the observations are positive we have the relationship among the three means given as: $\bar{x} \geq \text{GM} \geq \text{HM}$

ii. For two observations $\sqrt{\bar{x} * \text{HM}} = \text{GM}$

iii. $\bar{x} = \text{GM} = \text{HM}$ if all observation are positive and have equal value.

3.4.2. Median

Another useful measure of central tendency is the median. If the observations in the data set are arranged in increasing or decreasing order, the median is the middle observation, which divides the set into equal halves. If the number of observations n is odd, there will be a unique median, the $\tilde{x} = \left(\frac{n+1}{2}\right)^{\text{th}}$ number from either end in the ordered sequence. If n is even, there is strictly no middle observation, but the median is defined by convention as the average of the two middle observations, the $\tilde{x} = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{value} + \left(\frac{n}{2}+1\right)^{\text{th}} \text{value}}{2}$ from either end. This is used only the data are presented in the form of series.

Example: The two data sets: 8, 5, 4, 12, 15, 7, 28 and 8, 5, 4, 12, 15, 7, 49, for example, have different means but the same median, 8.

Median for ungrouped frequency distribution:- In this case also, the median is obtained by the above formula. After arranging the values in an increasing order find the smallest CF greater than

or equal to that value obtained by *odd&even* above formula and the corresponding value is the median.

Median for grouped frequency distribution:-For continuous data, the median is obtained by the following formula.

$$\text{Median} = L + \frac{w}{f_{\text{med}}} \left(\frac{n}{2} - CF \right) = \tilde{x}$$

Where: L = the lower class boundary of the median class; w = the class width of the median class;

f_{med} = the frequency of the median class; and CF = the cum. freq. corresponding to the class preceding the median class. That is, the sums of the frequencies of all classes lower than the median class. Where the median class is the class which contains the $(n/2)^{\text{th}}$ observation whether n is odd or even, since the items have already lost their originality once they are grouped in to continuous classes.

Example: Find the median amount of sulfur oxide emitted by the industrial plant table 3.3:

Table 3.3. Data for tons of sulfur oxide

Tons of sulfur oxide	Frequency (f_i)
5.0 – 8.9	3
9.0 – 12.9	10
13.0 – 16.9	14
17.0 – 20.9	25
21.0 – 24.9	17
25.0 – 28.9	9
29.0 – 32.9	2
<i>Total</i>	<i>80</i>

Solution: Since $N = 80$, $N/2 = 40$, the median is located in the 16.95 – 20.95 class interval.

Hence $L_1 = 16.95$, $f_{\text{median}} = 25$, $\sum f_1 = 3+10+14 = 27$, and $C = 4$.

$$\tilde{x} = 16.95 + \left(\frac{40 - 27}{25} \right) (4) = 19.0$$

Remark: For symmetric distributions, the arithmetic mean will be approximately the same as the median. For positively skewed distributions, the arithmetic mean will tend to be

larger than the median; for negatively skewed distributions, the arithmetic mean will tend to be smaller than the median.

Example: Calculate the median for the following frequency distribution

C.I	1 - 5	6 - 10	11 - 15	16 - 20	21 - 25	26 - 30	31 - 35	Total
Freq.	4	8	12	6	3	4	3	40

Solution: Construct the less than cumulative frequency distribution, then

C.I	1 - 5	6 - 10	11 - 15	16 - 20	21 - 25	26 - 30	31 - 35	Total
Freq.	4	8	12	6	3	4	3	40
Cuml. Freq.	4	12	24	30	33	37	40	

Since $n = 40$, $40/2 = 20$, and the smallest CF greater than or equal to 20 is 24; thus, the median class

is the third class. And for this class, $L = 10.5$, $w = 5$, $f_{med} = 12$, $CF = 12$. Then applying the formula,

We get:

$$\tilde{x} = 10.5 + (20 - 12) * 5 / 12 = 13.8$$

Therefore, the advantage of the median as a measure of location is that it is less affected by extreme observations. However, the median has some disadvantages in comparison with the mean:

1. It takes no account of the precise magnitude of most of the observations and is therefore less efficient than the mean because it wastes information.
2. If two groups of observations are pooled, the median of the combined group cannot be expressed in terms of the medians of the two component groups. However, the mean can be so expressed.

3.4.3. Mode

A third measure of central tendency is the mode. It is the value at which the frequency polygon reaches a peak. The mode is not used widely in analytical statistics, other than as a descriptive measure, mainly because of the ambiguity in its definition, as the fluctuations of small frequencies are apt to produce spurious modes. For these reasons, in the remainder of the book we focus on a single measure of location, the mean.

The mode mayn't exist, and even if it does exist it mayn't be unique. *Uni-modal*: is a distribution having one mode. *Bimodal*: is a distribution with two modes. Multimodal is a distribution having three or more modes.

Example: the modal age of the age distribution: 23, 28, 28, 31, 32, 34, 37, 42, 50, and 61 is 28, since it occurred twice while the other values occurred only once.

For uni-modal frequency curves which are moderately asymmetrical, we have the empirical relation:

$$\text{Mean} - \text{mode} = 3(\text{Mean} - \text{Median})$$

The Mode for Grouped Data

When designing the mode of grouped data, we usually refer to the modal class interval, where the modal class is the interval with the highest frequency. From a frequency distribution or histogram the mode can be obtained from the formula:

$$\text{Mode} = (\hat{X}) = L_1 + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \cdot C$$

Where L_1 = Lower class boundary of modal class

Δ_1 = Excess of modal frequency over frequency of next lower class

Δ_2 = Excess of modal frequency over frequency of next higher class

C = Size of modal class interval

Example: Find the modal amount of sulfur oxide emitted by the industrial plant given table 3.4.

Table 3.4: frequency distribution for sulfur oxide emitted

Tons of sulfur oxide	Frequencies
5.0 – 8.9	3
9.0 – 12.9	10
13.0 – 16.9	14

17.0 – 20.9	25
21.0 – 24.9	17
25.0 – 28.9	9
29.0 – 32.9	2

Solution: The modal class is 17.0 – 20.9, since it has the highest frequency. Hence $L_1=16.95$, $\Delta_1 = 25 - 14 = 11$, $\Delta_2=25 - 17 = 8$, $C = 4$.

Then,

$$\hat{X} = 16.95 + [11/(11+8)](4) = 19.27$$

Chapter-4

4. Measures of Variation (Dispersion), Skewness and Kurtosis

In the preceding sections several measures which are used to describe the central tendency of a distribution were considered. While the mean, median, etc. give useful information about the center of the data, we also need to know how “spread out” the numbers are about the center.

4.1 Objectives of measures of central tendency

Just as central tendency can be measured by a number in the form of an average, the amount of variation (dispersion, spread, or scatter) among the values in the data set can also be measured. The measures of central tendency describe that the major part of values in the data set appears to concentrate around a central value called average with the remaining values scattered (distributed) on either sides of that value. But these measures do not reveal how these values are dispersed (spread or scatter) on each side of the central value. The dispersion of values is indicated by the extent to which these values tend to spread over an interval rather than cluster closely around an average.

The term dispersion is generally used in two senses. Firstly, dispersion refers to the variations of the items among themselves. If the value of all the items of a series is the same, there will be no variation among different items of a series. Secondly, dispersion refers to the variation of the items around an average. If the difference between the value of items and the average is large, the dispersion will be high and on the other hand if the difference between the value of the items and averaging is small, the dispersion will be low. Thus, dispersion is defined as scatteredness or spreadness of the individual items in a given series.

After studying this chapter, you should be able to:

- ✓ Explain the meaning of measures of dispersion
- ✓ Compare two or more sets of data using relative measures of dispersion.
- ✓ Apply the Z-score to find out the relative standing of values.

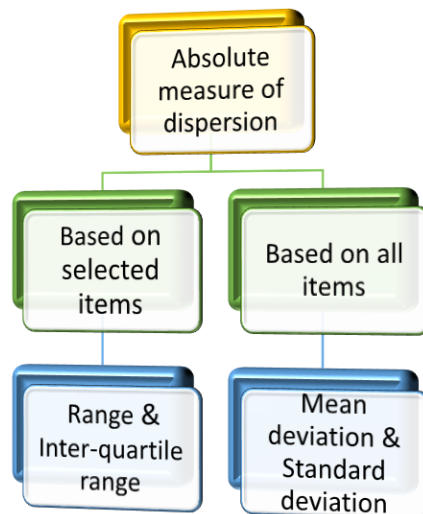
- ✓ Explain measures of skewness and kurtosis.

Objectives of measuring Variation:

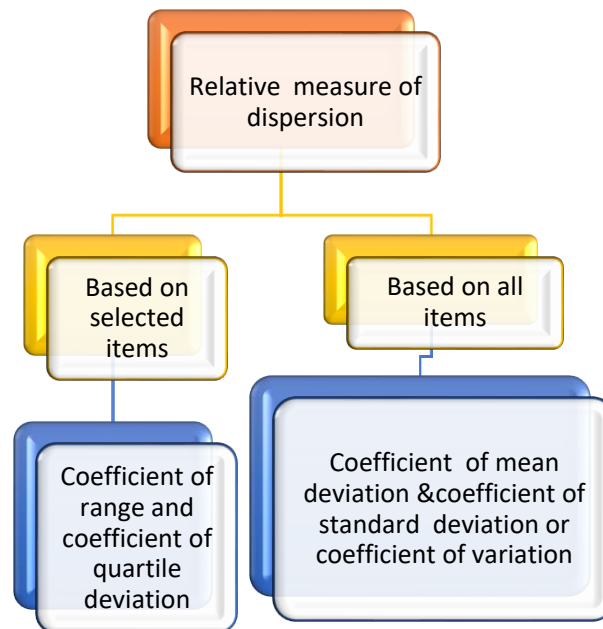
- ✓ To judge the reliability of measures of central tendency
- ✓ To control variability itself.
- ✓ To compare two or more groups of numbers in terms of their variability.
- ✓ To make further statistical analysis.

4.2 Absolute and Relative Measures of Dispersion

Absolute measures of dispersion: Absolute measure is expressed in the same statistical unit in which the original data are given such as kilograms, tones etc. These measures are suitable for comparing the variability in two distributions having variables expressed in the same units and of the same averaging size. These measures are not suitable for comparing the variability in two distributions having variables expressed in different units.



Relative measures of dispersion: A relative measure of dispersion is the ratio of a measure of absolute dispersion to an appropriate average or the selected items of the data.



4.3 Types of Measures of Variation

4.3.1 The Range and Relative Range

The range is defined as the difference between the highest and smallest observation in the data. It is the crudest measure of dispersion. The range is a measure of absolute dispersion and as such cannot be usefully employed for comparing the variability of two distributions expressed in different units.

Range is the simplest measures of dispersion. It is defined as the difference between the largest and smallest value in a given set of data. Its formula is:

$$R = L - S$$

Where R=Range, L= Largest value in a given set of data, S= smallest value in a given set of data.

For a **continuous grouped distribution**, the range may be obtained as:

- The difference between upper class limit of the last class and the lower class limit of the first class, or
- The difference between the largest class mark and the smallest class mark, or

- The difference between the upper class boundary of the last class and the lower class boundary of the first class.

The range is used in describing like the maximum change in daily temperature, rainfall, etc. When the sample size is small, it can be an adequate measure of variation. It is commonly used in quality control.

The relative measures of range, also called *coefficient of range*, is defined as

$$\text{Relative Range}(RR) = \frac{L - S}{L + S}$$

Example 4.1: Five students obtained the following marks in statistics: 20, 35, 25, 30, 15. Find the range and relative range

Solution: Here, $L = 35$, and $S = 15$

$$\text{Range} = L - S = 35 - 15 = 20$$

$$RR = \frac{L - S}{L + S} = \frac{35 - 15}{35 + 15} = 0.4$$

Example 4.2: Find out range and relative range of the following given data.

Size	5-10	11-15	16-20	21-25	26-30
Frequency	4	9	15	30	40

Solution: Here,

L = Upper class limit of the largest class = 30

S = lower class limit of the smallest class = 5

$$\text{Range} = 30 - 5 = 25, \quad RR = \frac{30 - 5}{30 + 5} = 0.7143 .$$

Merits of the Range

- It is well-defined, easy to compute and simple to understand.
- It helps in giving an idea about the variation, just by giving the lowest value and the greatest value of variable.

Demerits of the Range

- It is not based on all observations of the series.
- It can't be calculated in case of open-ended distribution.
- It is affected by sampling fluctuation.
- It is affected by extreme values in the series.

4.3.4 The Variance, Standard Deviation and Coefficient of Variation

Variance and Standard Deviation

Like the mean deviation, the variance is also based on all observations in a set of data. But the variance is the average of squared deviations from the mean. Recall that the sum of squared deviations is minimum only when taken from the mean. Squared deviations are mathematically manipulated than absolute deviations. Thus, if we averaged the squared deviations from the mean and take the square root of the result (to compensate for the fact that the deviations were squared), we obtain the standard deviation. This overcomes the limitation of the mean deviation.

Population Variance (σ^2)

If we divide the variation by the number of values in the population, we get something called the population variance. This variance is the "average squared deviation from the mean".

- **For ungrouped data (individual series)**

$$\sigma^2 = \frac{\sum_{i=1}^N (X_i - \mu)^2}{N} = \frac{1}{N} [\sum_{i=1}^N X_i^2 - N\mu^2]$$
 where μ is the population arithmetic mean and N is the total number of observations in the population.

- **For discrete data arranged in FD & for continuous grouped data**

$$\sigma^2 = \frac{\sum f_i (X_i - \mu)^2}{N} = \frac{1}{N} [\sum f_i X_i^2 - N\mu^2]$$
 where μ is the population arithmetic mean, X_i is the class mark of the i^{th} class, f_i is the frequency of the i^{th} class and $N = \sum f_i$

Sample Variance (S^2)

One would expect the sample variance to simply be the population variance with the population mean replaced by the sample mean. However, one of the major uses of statistics is to estimate the corresponding parameter. This formula has the problem that the estimated value isn't the same as the parameter. To offset this, the sum of the squares of the deviations is divided by one less than the sample size.

- **For ungrouped data**

$$S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} = \frac{1}{n-1} [\sum_{i=1}^n x_i^2 - n\bar{x}^2]$$
 where $\bar{x}$ is the sample arithmetic mean and n is the total number of observations in the sample.

- **For discrete data arranged in FD**

If the values x_i have frequencies f_i ($i=1,2,\dots,m$), then the sample variance is given by:

$$S^2 = \frac{\sum f_i(x_i - \bar{x})^2}{n-1} = \frac{1}{n-1} [\sum f_i x_i^2 - n\bar{x}^2] \quad \text{or} \quad S^2 = \frac{1}{n-1} \sum_{i=1}^m f_i (x_i - \bar{x})^2$$

- **For continuous grouped data**

$S^2 = \frac{\sum f_i(x_i - \bar{x})^2}{n-1} = \frac{1}{n-1} [\sum f_i x_i^2 - n\bar{x}^2]$ where $\bar{x}$ is the sample arithmetic mean, x_i is the class mark of the i^{th} class, f_i is the frequency of the i^{th} class and $n = \sum f_i$.

The Standard Deviation

There is a problem with variances. Recall that the deviations were squared. That means that the units were also squared. To get the units back the same as the original data values, the square root must be taken.

➤ Population Standard Deviation (σ)

$\sigma = \sqrt{\sigma^2}$ where σ^2 is the population variance.

➤ Sample Standard Deviation (S)

$S = \sqrt{S^2}$ where S^2 is the sample variance.

Example 4.7: Find the sample variance and standard deviation of:

x_i	2	4	5	6	8
f_i	2	2	3	1	2

Solution: Prepare the following table:

x_i	f_i	$f_i x_i$	x_i^2	$f_i x_i^2$
2	2	4	4	8
4	2	8	16	32
5	3	15	25	75
6	1	6	36	36
8	2	16	64	128
Sum	10	49		279

Thus, $n = \sum f_i = 10$, $\sum f_i x_i = 49$, $\sum f_i x_i^2 = 279$.

$$S^2 = \frac{1}{n-1} \left[\sum f_i x_i^2 - n \bar{x}^2 \right]$$

$$= \frac{1}{9} \left[279 - 10 \left(\frac{49}{10} \right)^2 \right] = \frac{1}{9} (38.9) = 4.32, \text{ and } S = \sqrt{4.32} = 2.08.$$

Example 4.8: Find the sample variance and standard deviation for the distribution:

C.I	1-5	6-10	11-15	16-20
Freq.	4	1	2	3

Solution: In a continuous F.D., x_i is the class mark representing the i^{th} class.

C.I	x_i	f_i	$f_i x_i$	$f_i x_i^2$
1-5	3	4	12	36
6-10	8	1	8	64
11-15	13	2	26	338
16-20	18	3	54	972
Total		10	100	1410

Where, $n = \sum f_i = 10$, $\bar{x} = \frac{\sum f_i x_i}{n} = \frac{100}{10} = 10$, $\sum f_i x_i^2 = 1410$, so that

$$S^2 = \frac{1}{n-1} \left[\sum f_i x_i^2 - n \bar{x}^2 \right] = \frac{1}{9} [1410 - 10(10)^2]$$

$$= \frac{410}{9} = 45.56,$$

$$S = \sqrt{45.56} = 6.75.$$

Properties of Variance & Standard Deviation

1. If a constant is added to (or subtracted from) all the values, the variance remains the same; i.e., for any constant k , $V(x_i \pm k) = V(x_i)$.

Example 4.9 Consider the 6 sample values x_i : 54, 52, 53, 50, 51, and 52.

The sample variance is $2 = V(x_i)$. Now, subtract 50 from each value to get:

y_i : 4, 2, 3, 0, 1, 2; and, the variance of this new series is 2. i.e., $V(x) = V(y) = 2$.

2. If each and every value is multiplied by a non-zero constant (k), the standard deviation is multiplied by $|k|$ and the variance is multiplied by k^2 ; i.e., $V(kx_i) = k^2V(x_i)$.
3. Both the variance and the standard deviation give more weight to extreme values and less to those which are near to the mean.

Coefficient of Variation

The standard deviation is an absolute measure of deviation of observations around their mean and is expressed with the same unit of the data. Due to this nature of the standard deviation it is not directly used for comparison purposes with respect to variability. Therefore, it is useful to relate the arithmetic mean and SD together, since, for example, a standard deviation of 10 would mean something different conceptually if the arithmetic mean were 10 than if it were 1000. A special measure called the coefficient of variation, is often used for this purpose.

Definition: The coefficient of variation (CV) is defined by: $CV = \frac{\sigma}{\mu} \times 100\%$

The coefficient of variation is most useful in comparing the variability of several different samples, each with different means. This is because a higher variability is usually expected when the mean increases, and the CV is a measure that accounts for this variability. The coefficient of variation is also useful for comparing the reproducibility of different variables. CV is a relative measure free from unit of measurement. CV remains the same regardless of what units are used, because if the units are changed by a factor C , both the mean and SD change by the factor C ; the CV, which is the ratio between them, remains unchanged.

The standard deviation is an absolute measure of dispersion. The corresponding relative measure is known as the coefficient of variation (CV).

Of course, standard deviation is an absolute measure of dispersion that expresses the variation in the same unit as the original data but it can not be the sole basis for comparing two distributions. For instance, if we have a standard deviation of 10 and a mean of 5, the values vary by an amount twice as large as the mean itself. If, on the other hand, we have a standard deviation of 10 and a mean of 5000, the variation relative to the mean is significant. Therefore, we cannot know the dispersion of a set of data until we know the standard deviation, the mean, and how the standard deviation compares with the mean.

Coefficient of variation is used in such problems where we want to compare the variability of two or more different series. Coefficient of variation is the ratio of the standard deviation to the arithmetic mean, usually expressed in percent.

$$CV = \frac{\text{Standard deviation}}{\text{mean}} \times 100\%$$

For population data:

$$CV = \frac{\sigma}{\mu} \times 100$$

Where σ is the population standard deviation and μ is population mean.

For sample data:

$$CV = \frac{S}{\bar{x}} \times 100$$

Where S is the sample standard deviation and $\bar{x}$ is sample mean.

Remark: A distribution having less coefficient of variation is said to be less variable or more consistent or more uniform or more homogeneous.

Example 4.10: Last semester, the students of Mathematics and Chemistry Departments took *Introduction to Statistics* course. At the end of the semester, the following information was recorded.

Department	Mathematics	Chemistry
Mean score	85	65
Standard deviation	25	12

Compare the relative dispersions of the two departments' scores using the appropriate way.

Solution:

Mathematics Departments

$$\begin{aligned} CV &= \frac{S}{\bar{x}} \times 100 \\ &= \frac{25}{85} \times 100 \\ &= 29.41\% \end{aligned}$$

Chemistry Departments

$$\begin{aligned} CV &= \frac{S}{\bar{x}} \times 100 \\ &= \frac{12}{65} \times 100 \\ &= 18.46\% \end{aligned}$$

Interpretation: Since the CV of Mathematics Department students is greater than that of Chemistry Department students, we can say that there is more dispersion relative to the mean in the distribution of Mathematics students' scores compared with that of Chemistry students.

4.4 Standard Scores (Z-Scores)

A standard score for sample value in a data set is obtained by subtracting the mean of the data set from the value and dividing the result by the standard deviation of the data set. Basically, the standard score (z-score) tells us how many standard deviations a specific value is above or below the mean value of the data set. That is, the z-score is the number of standard deviations the data value falls above (positive z-score) or below (negative z-score) the mean for the data set.

Z-score computed from the population

$$Z \text{ score} = \frac{X - \mu}{\sigma}$$

Z-score computed from the sample

$$Z \text{ score} = \frac{X - \bar{X}}{S}$$

Example 4.11: What is the Z-score for the value of 14 in the following sample data set?

3 8 6 14 4 12 7 10

Solution:

$$\bar{X} = 8, \text{ SD} = 3.8173 \quad \text{thus, } Z = \frac{14-8}{3.8173} \approx 1.57.$$

∴ The data value of 14 is located 1.57 standard deviations above the mean 8 because the z-score is positive.

Example 4.12: Suppose that a student scored 66 in Statistics and 80 in Mathematics. The score of the summary of the courses is given below.

Course	Average score	Standard deviation of the score

Statistics	51	12
Mathematics	72	16

In which course did the student scored better as compared to his classmates?

Solution:

$$\text{Z-score of student in Statistics: } Z = \frac{X - \mu}{\sigma} = \frac{66 - 51}{12} = \frac{15}{12} = 1.25$$

$$\text{Z-score of student in Mathematics: } Z = \frac{X - \mu}{\sigma} = \frac{80 - 72}{16} = \frac{8}{16} = 0.5$$

From these two standard scores, we can conclude that the student has scored better in Statistics course relative to his classmates than in Mathematics course.

4.5 *Moments, Skewness and Kurtosis*

The measures of central tendency and variation discussed in previous one do not reveal the entire story about a frequency distribution. Two distributions may have the same mean and standard deviation but may differ in their shape of the distribution. Further description of their characteristics is necessary that is provided by measures of skewness and kurtosis.

4.5.1 Moments

Moments are statistical tools used in statistical investigation. The moments of a distribution are the arithmetic mean of the various powers of the deviations of items from some number. In our course, we shall use it in the study of *Skewness* and *Kurtosis* of statistical distribution.

Moments about the origin

$$M_r = \frac{\sum X_i^r}{n}$$

Where $r = 0, 1, 2, 3, \dots$

Moments about the origin for *grouped frequency distribution* and for *ungrouped frequency distribution* is

$$M_r = \frac{\sum f_i X_i^r}{n}$$

Where f_i is the frequency of X_i . X_i is the midpoint in the case of *grouped frequency distribution* or *class value* in the case of *ungrouped frequency distribution*.

Note that: $M_1 = \bar{X}$, $M_0 = 1$

Moments about the Mean (Central Moments)

$$M'_r = \frac{\sum (X_i - \bar{X})^r}{n}$$

Moments about the mean for *grouped frequency distribution* and for *ungrouped frequency distribution*.

$$M'_r = \frac{\sum f_i (X_i - \bar{X})^r}{n}$$

Where f_i is the frequency of X_i . X_i is the midpoint in the case of *grouped frequency distribution* or *class value* in the case of *ungrouped frequency distribution*.

Note that: $M'_2 = SD^2$ if it is assumed $n = n - 1$.

Moments about any arbitrary constant A

$$M'_r = \frac{\sum (X_i - A)^r}{n}$$

Moments about any arbitrary constant A for *grouped frequency distribution* and for *ungrouped frequency distribution*

$$M'_r = \frac{\sum f_i (X_i - A)^r}{n} .$$

Example 4.13: Find the first four moments about the mean for the following individual series

X_i : 3 6 8 10 18

Solution: $n=5$,

S.No	X_i	$(X_i - \bar{X})$	$(X_i - \bar{X})^2$	$(X_i - \bar{X})^3$	$(X_i - \bar{X})^4$
1	3	-6	36	-216	1296
2	6	-3	9	-27	81
3	8	-1	1	-1	1
4	10	1	1	1	1
5	18	9	81	729	6561
Total	$\sum X = 45$	$\sum (X - \bar{X}) = 0$	$\sum (X - \bar{X})^2 = 128$	$\sum (X - \bar{X})^3 = 486$	$\sum (X - \bar{X})^4 = 7940$

Thus,

$$\bar{X} = \frac{45}{5} = 9, M'_1 = \frac{\sum (X_i - 9)^1}{5} = 0, M'_2 = \frac{\sum (X_i - 9)^2}{5} = \frac{128}{5} = 25.6, M'_3 = \frac{\sum (X_i - 9)^3}{5} = \frac{486}{5} = 97.2$$

$$M'_4 = \frac{\sum (X_i - 9)^4}{5} = \frac{7940}{5} = 1588$$

4.5.2 Skewness

Skewness refers to lack of symmetry (or departure from symmetry) in a distribution.

- A skewed frequency distribution is one that is not symmetrical.
- Skewness is concerned with the shape of the curve not size.

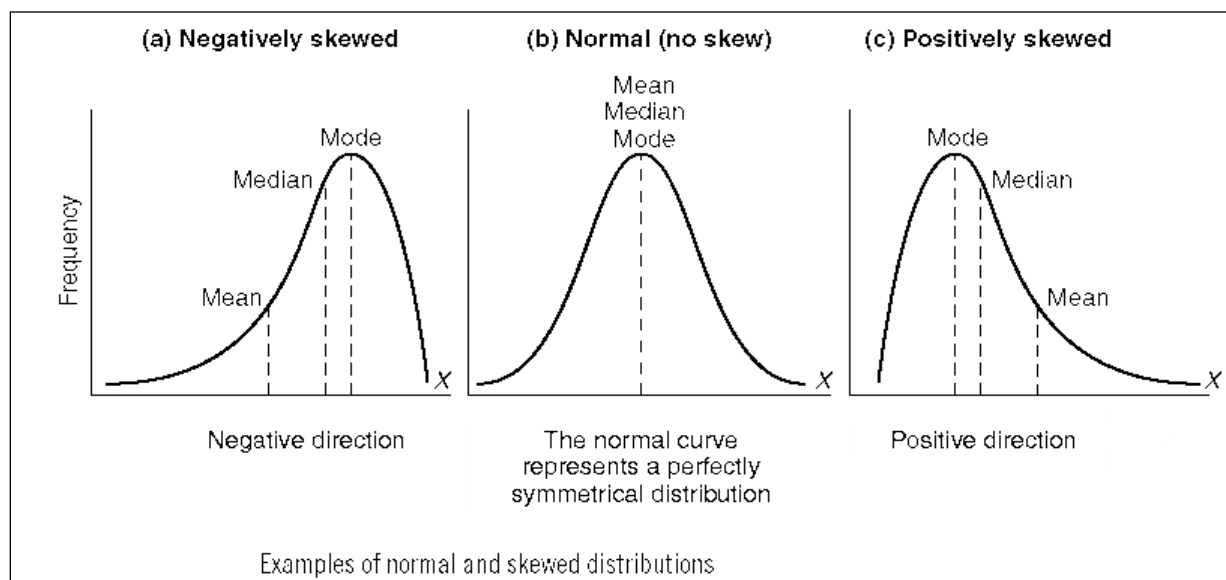
A distribution is said to be **symmetrical** when the value is uniformly distributed around the mean (distribution of the data below the mean and above the mean are equal). In a symmetrical distribution, the mean, median and mode coincide (i.e., mean = median = mode).

Positively skewed distribution: if the value of mean is greater than the mode, skewness is said to be positive. In a positively skewed distribution mean is greater than the mode and the median lies somewhere in between mean and mode. A positively skewed distribution contains some values that are much larger than the majority of other observations.

Negatively Skewed distribution: if the value of mode is greater than the mean, skewness is said to be negative. In a negatively skewed distribution mode is greater than the mean and the median lies in between mean and mode. The mean is pulled towards the low-valued item (that is, to the left). A negatively skewed distribution contains some values that are much smaller than the majority of observations.

Note that: In *moderately* skewed distributions the averages have the following relationship.

$$(\text{Mean} - \text{mode}) = 3(\text{mean} - \text{median})$$



How to check the presence of skewness in a distribution?

Skewness present in the data if:

- i) the graph is not symmetrical.
- ii) the mean, median and mode do not coincide.
- iii) the sum of positive and negative deviations from the median is not zero.
- iv) the frequencies are not similarly distributed on either side of the mode.

Measures of skewness (α_3)

A measure of skewness gives a numerical expression for and the direction of asymmetry in a distribution. It gives information about the shape of the distribution and the degree of variation on either side of the central value. The three most commonly used measures of skewness are *Pearson's coefficient of skewness*, *Bowley's coefficient of skewness* and *coefficient of skewness based on moments*.

1. Pearson's coefficient skewness (*Pearsonian coefficient of skewness*)

The skewness of the distribution can be measured by Pearson's Coefficient of Skewness (α_3), for which the formula is given below:

$$\alpha_3 = \frac{\text{Mean} - \text{Mode}}{\text{Standard deviation}}$$

2. Moment Coefficient of Skewness

Moment coefficient of skewness is based on *moments*. The formula for calculating coefficient of skewness is:

$$\alpha_3 = \frac{M'_3}{M'^{3/2}_2} = \frac{M'_3}{\sigma^3}$$

Where, $M'_r = \sum_{i=1}^n (x_i - \bar{x})^r / n$

The shape of the curve is determined by the value of α_3

$\alpha_3 > 0$, $\rightarrow$ the distribution is **positively** skewed/skewed to the **right**, i.e. **mode < median < mean**

$\rightarrow$ smaller observations are more frequent than larger observations. i.e., the majority of

the observations have a value below an average.

$\alpha_3 = 0$, $\rightarrow$ the distribution is **symmetric**, i.e. **mean = mode = median**

$\alpha_3 < 0$, $\rightarrow$ the distribution is **negatively** skewed/skewed to the **left**. i.e., **mean < median < mode**

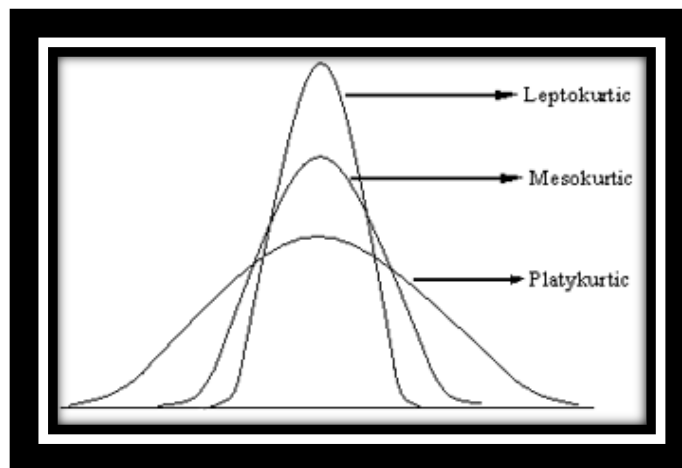
$\rightarrow$ smaller observations are less frequent than larger observations. i.e., the majority of

the observations have a value above an average.

4.4 Kurtosis

4.5.3 Kurtosis

Kurtosis is a measure of peakedness of a distribution. The degree of kurtosis of a distribution is measured relative to the peakedness of a normal curve. If a curve is more peaked than the normal curve it is called '*leptokurtic*'; if it is more or flate-topped than the normal curve it is called '*platykurtic*' or flat-topped. The normal curve itself is known as '*mesokurtic*'.

Measures of Kurtosis (α_4)

The moment coefficient of kurtosis:

$$\alpha_4 = \frac{M'_4}{M'^2_2} = \frac{M'_4}{\sigma^4}$$

The peakedness depends on the value of α_4

- $\alpha_4 > 3 \rightarrow$ the curve is leptokurtic,
- $\alpha_4 = 3 \rightarrow$ the curve is mesokurtic,
- $\alpha_4 < 3 \rightarrow$ the curve is platykurtic.

Example: Based on the following data:

$$M'_0 = 1, M'_1 = -0.6, M'_2 = 1.6, M'_3 = -2.4, M'_4 = 5.8$$

a/ Find the coefficient of skewness and discuss the distribution type.

b/ Find the coefficient of kurtosis and discuss the distribution type.

Solution:

$$\text{a/ } \alpha_3 = \frac{M'_3}{M'^{3/2}_2} = \frac{-2.4}{1.6^{3/2}} = -1.19 < 0, \rightarrow \text{the distribution is negatively skewed.}$$

$$\text{b/ } \alpha_4 = \frac{M'_4}{M'^2_2} = \frac{5.8}{1.6^2} = 2.26 < 3, \rightarrow \text{the curve is platykurtic.}$$

Example 4.14: Find the coefficient of skewness and the coefficient of kurtosis for the above example 4.13.

Solution:

$$\text{i) } \alpha_3 = \frac{M'_3}{M'^{3/2}_2} = \frac{97.2}{(25.6)^{3/2}} = \frac{97.2}{129.527} = 0.75$$

$\rightarrow$ the distribution is positively skewed.

$$\text{ii) } \alpha_4 = \frac{M'_4}{M'^2_2} = \frac{1588}{25.6^2} = 2.42$$

$\rightarrow$ the curve is platykurtic.

Exercise 4

1. Calculate the mean deviation about the mean, median and mode, and their coefficients and also variance and standard deviation for the following data.

Size of shoes	3	6	11	2	4	10	5	7	8	9
No. of pairs sold	10	15	25	6	4	3	2	8	9	4

2. An analysis of the monthly wages paid (in birr) to workers in two firms A and B belonging to the same industry gives the following results.

Value	Firm A	Firm B
Mean wage	52.5	47.5
Variance	100	121

In which firm A or B is there greater variability in individual wages?

3. A meteorologist interested in the consistency of temperatures in three cities during a given week collected the following data. The temperatures for the five days of the week in the three cities were

City 1: 25, 24, 23, 26, 17

City 2: 22, 21, 24, 22, 20

City 3: 32, 27, 35, 24, 28

Which city have the most consistent temperature, based on these data?

4. Some characteristics of annually family income distribution (in Birr) in two regions is as follows:

Region	Mean	Median	Standard deviation
A	6250	5100	960
B	6980	5500	940

- Calculate coefficient of skewness for each region
 - For which region the income is more consistent?
- The median and the mode of a mesokurtic distribution are 32 and 34 respectively. The 4th moment about the mean is 243. Compute the Pearsonian coefficient of skewness and identify the type of skewness. Assume $(n-1 = n)$.
 - If the standard deviation of a symmetric distribution is 10, what should be the value of the fourth moment so that the distribution is mesokurtic?